

Projectivity, Extension, and Square-Type Characterizations for Φ -Singular and Φ -Cosingular Operators

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Status

This is a likely-valid partial structural packet for the questions in Gonzalez–Salas–Brown, *Two classes of operators related to the perturbation classes problem*, arXiv:2302.04506. It does not prove the global equalities $\mathcal{SS} = \Phi\mathcal{S}$ or $\mathcal{SC} = \Phi\mathcal{C}$. It gives two structural advances and two sharp localization theorems:

reflexive duality turns the source paper’s one-way adjoint results into equivalences,

$$Y \times Y \hookrightarrow Y \text{ forces } \Phi\mathcal{S}(X, Y) = P\Phi_+(X, Y),$$

$$X \twoheadrightarrow X \times X \text{ forces } \Phi\mathcal{C}(X, Y) = P\Phi_-(X, Y),$$

subprojective/superprojective operators cannot witness strictness of $\mathcal{SS} \subset \Phi\mathcal{S}$ or $\mathcal{SC} \subset \Phi\mathcal{C}$,

extension/lifting properties force the remaining perturbation classes back to $\mathcal{SS}$ and $\mathcal{SC}$.

Source Question

The source paper asks whether the new classes $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$ are subspaces, operator ideals, or satisfy the same multiplication stability as the perturbation classes, and asks whether the inclusions $\mathcal{SS}(X, Y) \subset \Phi\mathcal{S}(X, Y)$ and $\mathcal{SC}(X, Y) \subset \Phi\mathcal{C}(X, Y)$ can be strict.

We state some basic questions on the class $\Phi\mathcal{S}$.

Question 1. *Suppose that $\Phi_+(X, Y) \neq \emptyset$.*

- (a) *Is $\Phi\mathcal{S}(X, Y)$ a subspace of $\mathcal{L}(X, Y)$?*
- (c) *Is $\Phi\mathcal{S}$ an operator ideal?*
- (c) *Is Proposition 2.2 valid for $\Phi\mathcal{S}$?*

Answering a question in [7], an example of an operator $K \in P\Phi_+ \setminus \Phi\mathcal{S}$ was given in [3, Example 2.3], but we do not know if the other inclusion can be strict.

Question 2. *Suppose that $\Phi_+(X, Y) \neq \emptyset$. Is $\mathcal{SS}(X, Y) = \Phi\mathcal{S}(X, Y)$?*

A negative answer to Question 2 would provide a new counterexample to the perturbation classes problem for Φ_+ .

Let us see that the inclusions in Proposition 3.2 become equalities in some cases.

An infinite dimensional Banach space Y is isomorphic to its square, denoted $Y \times Y \simeq Y$, in many cases: $L_p(\mu)$ and ℓ_p ($1 \leq p \leq \infty$), c_0 , and $C[0, 1]$. On the other hand, James' space J and some spaces of continuous functions on a compact like $C[0, \omega_1]$ are not isomorphic to their square, where ω_1 is the first uncountable ordinal. See [5] and [23].

Theorem 3.4. *Suppose that the spaces Y and V satisfy $\Phi_-(Y \times V) \neq \emptyset$.*

The corresponding $\Phi\mathcal{C}$ questions are:

$[S, I_N] \in \Psi_-(X \times X, Y)$. Hence $I_N \in \Psi\mathcal{C}(X, Y)$ implies $S \in \Psi_-(X, Y)$. $\square$

Question 4. *Suppose that $\Phi_-(X, Y) \neq \emptyset$.*

- (a) *Is $\Phi\mathcal{C}(X, Y)$ a subspace of $\mathcal{L}(X, Y)$?*
- (c) *Is $\Phi\mathcal{C}$ an operator ideal?*
- (c) *Is Proposition 2.2 valid for $\Phi\mathcal{C}$?*

Answering a question in [7], an example of an operator $K \in P\Phi_- \setminus \Phi\mathcal{C}$ was given in [3], but we do not know if the other inclusion can be strict.

Question 5. *Suppose that $\Phi_-(X, Y) \neq \emptyset$. Is $\mathcal{SC}(X, Y) = \Phi\mathcal{C}(X, Y)$?*

A negative answer to Question 5 would provide a new counterexample to the perturbation classes problem for Φ_- .

Next we will show that the inclusions in Proposition 4.2 become equalities in some cases.

Theorem 4.4. *Suppose that the spaces Y and V satisfy $\Phi_-(Y \times V) \neq \emptyset$.*

Later in the paper, Proposition 4.9 proves the one-way implications

$$K^* \in \Phi\mathcal{S}(Y^*, X^*) \Rightarrow K \in \Phi\mathcal{C}(X, Y), \quad K^* \in \Phi\mathcal{C}(Y^*, X^*) \Rightarrow K \in \Phi\mathcal{S}(X, Y).$$

$$= \langle S^*x + T^*x, x \rangle = \langle (S^* + T^*)x, x \rangle.$$

As a consequence, $[S, T] \in \Phi_-$ if and only if $(S^*, T^*) \in \Phi_+$. Similarly, $(S, T)^*$ can be identified with $[S^*, T^*]$.

The following result describes the behavior of the classes of Φ -singular and Φ -cosingular operators under duality.

Proposition 4.9. *Let $K \in \mathcal{L}(X, Y)$.*

- (1) *If $K^* \in \Phi\mathcal{S}(Y^*, X^*)$ then $K \in \Phi\mathcal{C}(X, Y)$.*
- (2) *If $K^* \in \Phi\mathcal{C}(Y^*, X^*)$ then $K \in \Phi\mathcal{S}(X, Y)$.*

Proof. (1) Let $S \in \mathcal{L}(X, Y)$ such that $[S, K] \in \Phi_-(X \times X, Y)$. Then $[S, K]^* \in \Phi_+$. Since $[S, K]^* \equiv (S^*, K^*)$, we have $(S^*, K^*) \in \Phi_+(Y^*, X^* \times X^*)$, and from $K^* \in \Phi\mathcal{S}(Y^*, X^*)$ we obtain $S^* \in \Phi_+(Y^*, X^*)$; therefore $S \in \Phi_-(X, Y)$, and hence $K \in \Phi\mathcal{C}(X, Y)$.

Definitions

For $S, K \in \mathcal{L}(X, Y)$, write

$$(S, K)x = (Sx, Kx) \in Y \times Y, \quad [S, K](x_1, x_2) = Sx_1 + Kx_2 \in Y.$$

Assume the corresponding semi-Fredholm classes are nonempty. An operator $K \in \mathcal{L}(X, Y)$ is Φ -singular, $K \in \Phi\mathcal{S}(X, Y)$, if

$$(S, K) \in \Phi_+(X, Y \times Y) \implies S \in \Phi_+(X, Y)$$

for every $S \in \mathcal{L}(X, Y)$. It is Φ -cosingular, $K \in \Phi\mathcal{C}(X, Y)$, if

$$[S, K] \in \Phi_-(X \times X, Y) \implies S \in \Phi_-(X, Y)$$

for every $S \in \mathcal{L}(X, Y)$.

We also use the operator versions of subprojectivity and superprojectivity from [3]. An operator $T \in \mathcal{L}(X, Y)$ is subprojective if every closed infinite-dimensional subspace $M \subset X$ on which T is an isomorphism contains a closed infinite-dimensional subspace N such that $T(N)$ is complemented in Y . Dually, T is superprojective if every closed infinite-codimensional subspace $M \subset Y$ for which $Q_M T$ is onto is contained in a closed infinite-codimensional subspace $N \subset Y$ such that $T^{-1}(N)$ is complemented in X .

We say that the pair (Y, Y) has the qualitative linear extension property if every bounded operator $R : E \rightarrow Y$, defined on a closed subspace $E \subset Y$, has a bounded extension $\tilde{R} : Y \rightarrow Y$. This is the qualitative version of the standard linear extension property for pairs of Banach spaces [4]. We say that X has the lifting property if, for every quotient map $q : E \rightarrow F$ and every $A \in \mathcal{L}(X, F)$, there is $T \in \mathcal{L}(X, E)$ such that $qT = A$.

Proof Intuition

There are two mechanisms.

First, the source proof dualizes $[S, K]$ and (S, K) . The only obstruction to reversing it is that an arbitrary operator $Y^* \rightarrow X^*$ need not be an adjoint. Reflexivity removes exactly that obstruction: every operator $A : Y^* \rightarrow X^*$ is S^* for $S = A^* : X \rightarrow Y$, after the canonical identifications $X = X^{**}$ and $Y = Y^{**}$.

Second, the square hypotheses let the new classes collapse to the usual perturbation classes. For $\Phi\mathcal{S}$, an embedding $U : Y \times Y \rightarrow Y$ converts a decoupled test (S, K) into a single operator $VS + WK$. If K belongs to the perturbation class $P\Phi_+$, then subtracting the perturbation WK leaves VS upper semi-Fredholm, and the injectivity of V gives $S \in \Phi_+$. For $\Phi\mathcal{C}$, a quotient map $Q : X \rightarrow X \times X$ pulls a decoupled lower semi-Fredholm test $[S, K]$ back to a single operator $SU_1 + KU_2$, and subtracting KU_2 forces S lower semi-Fredholm because U_1 is onto.

Third, subprojective and superprojective operators remove the main local escape route for a strict $\Phi\mathcal{S}$ or $\Phi\mathcal{C}$ counterexample. If a non-strictly singular operator is subprojective, then on some further infinite-dimensional subspace its image is complemented. This lets one subtract a bounded multiple of K from a fixed upper semi-Fredholm operator and create an infinite-dimensional kernel. The lower semi-Fredholm dual argument uses a complemented preimage of a large quotient kernel.

Fourth, the same subtraction trick works without complemented ranges when the ambient space itself supplies extensions or liftings. A perturbation-class operator K which is not strictly singular would make $A(K|_M)^{-1} : K(M) \rightarrow Y$ nonextendable for every upper semi-Fredholm A . Dually, a perturbation-class operator which is not strictly cosingular would force every lower semi-Fredholm test B to fail to lift through the quotient map $Q_M K$. Thus injective-type codomains and projective-type domains are too flexible to support strict witnesses.

Theorem 1: Reflexive Duality

Theorem 1. *Let X and Y be reflexive Banach spaces. Whenever the displayed classes are defined, for every $K \in \mathcal{L}(X, Y)$ one has*

$$K \in \Phi\mathcal{C}(X, Y) \iff K^* \in \Phi\mathcal{S}(Y^*, X^*)$$

and

$$K \in \Phi\mathcal{S}(X, Y) \iff K^* \in \Phi\mathcal{C}(Y^*, X^*).$$

Proof. The forward implications from the dual classes to the original classes are Proposition 4.9 of Gonzalez–Salas–Brown. We prove the converses.

Assume first that $K \in \Phi\mathcal{C}(X, Y)$. Let $A \in \mathcal{L}(Y^*, X^*)$ and suppose

$$(A, K^*) \in \Phi_+(Y^*, X^* \times X^*).$$

Since X and Y are reflexive, there is an operator $S \in \mathcal{L}(X, Y)$ such that $A = S^*$. Under the standard identification $[S, K]^* = (S^*, K^*)$, the preceding hypothesis says $[S, K]^* \in \Phi_+$. For any bounded operator T , $T^* \in \Phi_+$ if and only if $T \in \Phi_-$. Hence $[S, K] \in \Phi_-(X \times X, Y)$. Since $K \in \Phi\mathcal{C}(X, Y)$, it follows that $S \in \Phi_-(X, Y)$. Dualizing again gives $A = S^* \in \Phi_+(Y^*, X^*)$. Therefore $K^* \in \Phi\mathcal{S}(Y^*, X^*)$.

Now assume that $K \in \Phi\mathcal{S}(X, Y)$. Let $A \in \mathcal{L}(Y^*, X^*)$ and suppose

$$[A, K^*] \in \Phi_-(Y^* \times Y^*, X^*).$$

Again write $A = S^*$ for some $S \in \mathcal{L}(X, Y)$. Since $(S, K)^* = [S^*, K^*] = [A, K^*]$, the semi-Fredholm duality $T \in \Phi_+$ if and only if $T^* \in \Phi_-$ gives $(S, K) \in \Phi_+(X, Y \times Y)$. Since $K \in \Phi\mathcal{S}(X, Y)$, we have $S \in \Phi_+(X, Y)$. Hence $A = S^* \in \Phi_-(Y^*, X^*)$, proving $K^* \in \Phi\mathcal{C}(Y^*, X^*)$. $\square$

Corollary 1. *For reflexive X and Y , the structural questions for $\Phi\mathcal{S}(X, Y)$ are dual to the corresponding structural questions for $\Phi\mathcal{C}(Y^*, X^*)$. In particular, $\Phi\mathcal{S}(X, Y)$ is a linear subspace if and only if $\Phi\mathcal{C}(Y^*, X^*)$ is a linear subspace, and the strictness of $\mathcal{SS}(X, Y) \subset \Phi\mathcal{S}(X, Y)$ is equivalent, under adjoints, to the corresponding strictness question for $\Phi\mathcal{C}(Y^*, X^*)$ whenever the usual strictly singular/strictly cosingular adjoint duality applies in the spaces under consideration.*

Theorem 2: One-Sided Square Characterization

Theorem 2. *Let X and Y be Banach spaces, and assume the relevant classes are defined.*

1. *If $Y \times Y$ is isomorphic to a subspace of Y , then*

$$\Phi\mathcal{S}(X, Y) = P\Phi_+(X, Y).$$

2. *If $X \times X$ is isomorphic to a quotient of X , then*

$$\Phi\mathcal{C}(X, Y) = P\Phi_-(X, Y).$$

Proof. The inclusions $\Phi\mathcal{S}(X, Y) \subset P\Phi_+(X, Y)$ and $\Phi\mathcal{C}(X, Y) \subset P\Phi_-(X, Y)$ are Proposition 3.2 and Proposition 4.2 of the source paper. It remains to prove the reverse inclusions.

Suppose $Y \times Y$ embeds into Y . Let $U : Y \times Y \rightarrow Y$ be an into isomorphism, and write

$$U(y_1, y_2) = Vy_1 + Wy_2$$

with $V, W \in \mathcal{L}(Y)$. Let $K \in P\Phi_+(X, Y)$, and assume $(S, K) \in \Phi_+(X, Y \times Y)$. Since U is an into isomorphism,

$$U(S, K) = VS + WK \in \Phi_+(X, Y).$$

By Proposition 2.2 of the source paper, $WK \in P\Phi_+(X, Y)$. The class $P\Phi_+(X, Y)$ is a linear subspace, so $-WK \in P\Phi_+(X, Y)$. Applying the perturbation property to the upper semi-Fredholm operator $VS + WK$ gives

$$VS = (VS + WK) - WK \in \Phi_+(X, Y).$$

The operator $V : y \mapsto U(y, 0)$ is an into isomorphism, hence VS is upper semi-Fredholm if and only if S is upper semi-Fredholm. Therefore $K \in \Phi\mathcal{S}(X, Y)$.

Now suppose there is a quotient map $Q : X \rightarrow X \times X$, and write $Qx = (U_1x, U_2x)$. Since Q is onto, $U_1 : X \rightarrow X$ is onto. Let $K \in P\Phi_-(X, Y)$, and assume $[S, K] \in \Phi_-(X \times X, Y)$. Then

$$[S, K]Q = SU_1 + KU_2 \in \Phi_-(X, Y).$$

Here precomposition with a surjection preserves the range. By Proposition 2.2, $KU_2 \in P\Phi_-(X, Y)$, and hence

$$SU_1 = (SU_1 + KU_2) - KU_2 \in \Phi_-(X, Y).$$

Since U_1 is onto, $SU_1(X) = S(X)$; therefore S has finite-codimensional closed range exactly when SU_1 does. Thus $S \in \Phi_-(X, Y)$, and $K \in \Phi\mathcal{C}(X, Y)$. $\square$

Corollary 2. *Under the hypotheses of Theorem 2, the source paper's subspace and same-component multiplication-stability questions have positive answers. More precisely, $\Phi\mathcal{S}(X, Y)$ is a closed linear subspace and satisfies the componentwise stability of Proposition 2.2 whenever $Y \times Y \hookrightarrow Y$, while $\Phi\mathcal{C}(X, Y)$ has the analogous properties whenever $X \twoheadrightarrow X \times X$.*

Corollary 3. *In particular, the $\Phi\mathcal{S}$ characterization applies to every codomain Y which is isomorphic to its square, including the classical examples listed in the source paper such as ℓ_p , $L_p(\mu)$, c_0 , and $C[0, 1]$. The formulation also applies in the weaker situation where Y merely contains a closed subspace isomorphic to $Y \times Y$. Dually, the $\Phi\mathcal{C}$ characterization applies whenever X has a quotient isomorphic to $X \times X$, in particular whenever $X \simeq X \times X$.*

Theorem 3: Projective-Operator Barrier

Theorem 3. *Let X and Y be Banach spaces.*

1. *Suppose $\Phi_+(X, Y) \neq \emptyset$. If $K \in \mathcal{L}(X, Y)$ is a subprojective operator, then*

$$K \in P\Phi_+(X, Y) \iff K \in \Phi\mathcal{S}(X, Y) \iff K \in \mathcal{SS}(X, Y).$$

2. *Suppose $\Phi_-(X, Y) \neq \emptyset$. If $K \in \mathcal{L}(X, Y)$ is a superprojective operator, then*

$$K \in P\Phi_-(X, Y) \iff K \in \Phi\mathcal{C}(X, Y) \iff K \in \mathcal{SC}(X, Y).$$

Proof. We prove the first assertion. The inclusions $\mathcal{SS}(X, Y) \subset \Phi\mathcal{S}(X, Y) \subset P\Phi_+(X, Y)$ are from the source paper, so it is enough to show that a subprojective $K \in P\Phi_+(X, Y)$ is strictly singular.

Assume, to the contrary, that K is not strictly singular. There is a closed infinite-dimensional subspace $M \subset X$ such that $K|_M$ is an isomorphism. Since K is subprojective, M contains a closed infinite-dimensional subspace N such that $K(N)$ is complemented in Y . Let $P : Y \rightarrow K(N)$ be a bounded projection, and let

$$R = (K|_N)^{-1} : K(N) \rightarrow N.$$

Choose $A \in \Phi_+(X, Y)$, and define $L = ARP \in \mathcal{L}(Y)$. For every $n \in N$, $LKn = An$. Hence

$$S = A - LK$$

vanishes on N , so $S \notin \Phi_+(X, Y)$.

On the other hand, if $K \in P\Phi_+(X, Y)$, then Proposition 2.2 of the source paper gives $LK \in P\Phi_+(X, Y)$, and since the perturbation class is a linear subspace, also $-LK \in P\Phi_+(X, Y)$. Applying this perturbation to the upper semi-Fredholm operator A gives

$$A - LK = S \in \Phi_+(X, Y),$$

a contradiction. Therefore a subprojective $K \in P\Phi_+(X, Y)$ must be strictly singular.

For the second assertion, it is enough to show that a superprojective $K \in P\Phi_-(X, Y)$ is strictly cosingular. Suppose not. Then there is a closed infinite-codimensional subspace $M \subset Y$ such that $Q_M K : X \rightarrow Y/M$ is onto. Since K is superprojective, there is a closed infinite-codimensional subspace $N \subset Y$, with $M \subset N$, such that

$$E = K^{-1}(N)$$

is complemented in X . The map $Q_N K : X \rightarrow Y/N$ is onto and has kernel E . Because E is complemented, $Q_N K$ has a bounded right inverse $R : Y/N \rightarrow X$.

Choose $B \in \Phi_-(X, Y)$, and set

$$T = RQ_N B \in \mathcal{L}(X).$$

Then $Q_N KT = Q_N B$, so $S = B - KT$ has range contained in N . Since N has infinite codimension in Y , $S \notin \Phi_-(X, Y)$. But if $K \in P\Phi_-(X, Y)$, Proposition 2.2 gives $KT \in P\Phi_-(X, Y)$, and hence $-KT \in P\Phi_-(X, Y)$. Applying this perturbation to $B \in \Phi_-(X, Y)$ gives

$$B - KT = S \in \Phi_-(X, Y),$$

again a contradiction. Thus a superprojective $K \in P\Phi_-(X, Y)$ must be strictly cosingular. $\square$

Corollary 4. *Any counterexample to $\mathcal{SS}(X, Y) = \Phi\mathcal{S}(X, Y)$ must be a Φ -singular operator which is not subprojective in the sense of Gonzalez–Pello. Any counterexample to $\mathcal{SC}(X, Y) = \Phi\mathcal{C}(X, Y)$ must be a Φ -cosingular operator which is not superprojective.*

Consequently, $\mathcal{SS}(X, Y) = \Phi\mathcal{S}(X, Y)$ whenever every operator $X \rightarrow Y$ is subprojective, for instance when Y is subprojective [3, Corollary 2.4]. Similarly, $\mathcal{SC}(X, Y) = \Phi\mathcal{C}(X, Y)$ whenever every operator $X \rightarrow Y$ is superprojective, for instance when X is superprojective [3, Corollary 2.4].

Theorem 4: Extension–Lifting Obstruction

Theorem 4. *Let X and Y be Banach spaces.*

1. *Suppose $\Phi_+(X, Y) \neq \emptyset$. Let $K \in P\Phi_+(X, Y)$. If K is not strictly singular, then for every closed infinite-dimensional subspace $M \subset X$ on which $K|_M$ is an isomorphism and every $A \in \Phi_+(X, Y)$, the operator*

$$A|_M(K|_M)^{-1} : K(M) \longrightarrow Y$$

has no bounded extension to an operator $Y \rightarrow Y$. Consequently, if the pair (Y, Y) has the qualitative linear extension property, then

$$P\Phi_+(X, Y) = \Phi\mathcal{S}(X, Y) = \mathcal{SS}(X, Y).$$

2. *Suppose $\Phi_-(X, Y) \neq \emptyset$. Let $K \in P\Phi_-(X, Y)$. If K is not strictly cosingular, then for every closed infinite-codimensional subspace $M \subset Y$ for which $Q_M K : X \rightarrow Y/M$ is onto and every $B \in \Phi_-(X, Y)$, the operator*

$$Q_M B : X \longrightarrow Y/M$$

does not lift through $Q_M K$; equivalently, there is no $T \in \mathcal{L}(X)$ with

$$Q_M K T = Q_M B.$$

Consequently, if X has the lifting property, then

$$P\Phi_-(X, Y) = \Phi\mathcal{C}(X, Y) = \mathcal{SC}(X, Y).$$

Proof. We prove the upper semi-Fredholm assertion first. Let $K \in P\Phi_+(X, Y)$, and suppose that $M \subset X$ is closed infinite-dimensional and that $K|_M$ is an isomorphism. Then $K(M)$ is closed in Y . Choose $A \in \Phi_+(X, Y)$. If the operator

$$R = A|_M(K|_M)^{-1} : K(M) \rightarrow Y$$

extended to some $L \in \mathcal{L}(Y)$, then

$$S = A - LK$$

would vanish on M . Hence $S \notin \Phi_+(X, Y)$, because its kernel is infinite-dimensional. On the other hand, Proposition 2.2 of the source paper gives $LK \in P\Phi_+(X, Y)$; since the perturbation class is a linear subspace, $-LK \in P\Phi_+(X, Y)$. Perturbing the upper semi-Fredholm operator A by $-LK$ gives $A - LK = S \in \Phi_+(X, Y)$, a contradiction.

If (Y, Y) has the qualitative linear extension property and $K \in P\Phi_+(X, Y)$ is not strictly singular, then such an M exists and the preceding obstruction is impossible. Thus $P\Phi_+(X, Y) \subset \mathcal{SS}(X, Y)$. The reverse chain

$$\mathcal{SS}(X, Y) \subset \Phi\mathcal{S}(X, Y) \subset P\Phi_+(X, Y)$$

is Proposition 3.2 of the source paper, so equality follows.

For the lower semi-Fredholm assertion, let $K \in P\Phi_-(X, Y)$, and suppose that $M \subset Y$ is closed infinite-codimensional and that $q = Q_M K : X \rightarrow Y/M$ is onto. Choose $B \in \Phi_-(X, Y)$. If there were $T \in \mathcal{L}(X)$ with $qT = Q_M B$, then

$$S = B - KT$$

would have range contained in M . Since M has infinite codimension, $S \notin \Phi_-(X, Y)$. But Proposition 2.2 gives $KT \in P\Phi_-(X, Y)$, and hence $-KT \in P\Phi_-(X, Y)$. Perturbing $B \in \Phi_-(X, Y)$ by $-KT$ gives $B - KT = S \in \Phi_-(X, Y)$, a contradiction.

If X has the lifting property and $K \in P\Phi_-(X, Y)$ is not strictly cosingular, then a quotient $q = Q_M K$ as above exists; applying the lifting property to the map $Q_M B : X \rightarrow Y/M$ contradicts the obstruction just proved. Thus $P\Phi_-(X, Y) \subset \mathcal{SC}(X, Y)$, and equality follows from the source paper's chain

$$\mathcal{SC}(X, Y) \subset \Phi\mathcal{C}(X, Y) \subset P\Phi_-(X, Y).$$

□

Corollary 5. *The strictness problem has the following additional localization. A strict witness in $\Phi\mathcal{S}(X, Y) \setminus \mathcal{SS}(X, Y)$ must force nonextension of the operators $A|_M(K|_M)^{-1} : K(M) \rightarrow Y$ described above. A strict witness in $\Phi\mathcal{C}(X, Y) \setminus \mathcal{SC}(X, Y)$ must force nonlifting of the quotient maps $Q_M B$ through $Q_M K$.*

In particular, injective codomains, and more generally codomains Y for which the pair (Y, Y) has the qualitative linear extension property, have

$$\mathcal{SS}(X, Y) = \Phi\mathcal{S}(X, Y)$$

for every X with $\Phi_+(X, Y) \neq \emptyset$. Dually, projective domains, and more generally domains with the lifting property, have

$$\mathcal{SC}(X, Y) = \Phi\mathcal{C}(X, Y)$$

for every Y with $\Phi_-(X, Y) \neq \emptyset$.

Limitations and Review Notes

The reflexivity assumption is essential for this proof, because it is used to represent an arbitrary $A : Y^* \rightarrow X^*$ as an adjoint S^* . Without such a pre-adjoint representation, Proposition 4.9 need not reverse by this argument.

These theorems do not settle whether $\mathcal{SS} = \Phi\mathcal{S}$ or whether $\mathcal{SC} = \Phi\mathcal{C}$. Theorem 2 isolates the one-sided square mechanism already visible in the source paper's combined perturbation-class arguments and promotes it to explicit characterizations of the intermediate classes $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$. In particular it strengthens the source paper's square-isomorphism characterizations of these intermediate classes to one-sided square hypotheses, and gives full positive answers to the source subspace and Proposition 2.2 stability questions in those cases. The global operator-ideal question, with arbitrary outside domains and codomains, remains beyond this packet. Theorem 3 is a sharp projectivity obstruction theorem: it does not rule out strictness, but it says that any strictness witness must avoid the subprojective/superprojective operator classes introduced in [3]. Theorem 4 adds an independent extension/lifting obstruction: even a non-subprojective or non-superprojective witness must also defeat the explicit extension or lifting tests in that theorem.

Bounded novelty check: the run indexes had no prior entry for arXiv 2302.04506 before this packet. Web searches for exact $\Phi\mathcal{S}/\Phi\mathcal{C}$ duality and stability phrases found only the source arXiv paper. Targeted searches for $\Phi\mathcal{S}/\Phi\mathcal{C}$ together with subprojective or superprojective operators found only the Gonzalez–Pello paper introducing those operator classes, not the triangular perturbation consequence above. Crossref metadata was checked for the older perturbation-class papers of Gonzalez [5] and Weis [6]; no later source surfaced that states these isolated one-sided square characterizations for $\Phi\mathcal{S}$ and $\Phi\mathcal{C}$. A final search for Φ -singular/ Φ -cosingular operators together with extension, lifting, injective, and projective Banach-space terminology found the source paper and background extension-property literature, but no statement of the extension–lifting obstruction in Theorem 4. The result is a structural partial answer pending human review.

References

References

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